

LUMINARA: Architects of the Lost Radiance

Game Design Document — Phase 1 Submission
Funobotz Discovery World 3D Hackathon

Field	Detail
Team	Nandha - ML
Build Target	Mobile AR (Primary) - Android .apk
Engine	Unity 6.6 (6000.6) · URP · AR Foundation 6.6.2
Date	September 17, 2026
Target Age/Grade	Ages 12+ (Middle School & High School) — Designed to bridge advanced spatial reasoning with fundamental physics.

1 · Game Title & Logline

LUMINARA: Architects of the Lost Radiance

A spatial-computing puzzle adventure where players deploy physical AR marker cards to unlock ancient gateways, then command a GPU-driven swarm of 15,000 luminous particles to reconstruct shattered engineering marvels buried deep inside a procedurally-lit cave system — all rendered in real-time on a custom boid compute pipeline.

Elevator pitch in one breath: You hold real cards, the world responds — light obeys you, stone remembers you, and a broken civilization rebuilds itself particle by particle under your hands.

2 · Theme & Category Alignment

Theme #2 — Adventure Quest World

Luminara drops the player into **Aethyra**, a fallen civilization whose infrastructure was powered entirely by captured light. The world is dead — not because of war, but because the light *left*. The player's quest is to journey through Aethyra's ruins, understand why the radiance collapsed, and physically restore it. This is not a combat game. It is a quest of **discovery, traversal, and restoration** — the beating heart of the Adventure Quest archetype.

Every location is a question: *What did this place used to be, and how do I bring it back?*

Category #4 — Engineering Puzzle Game

Every puzzle in Luminara is an **engineering problem**, not a riddle. Players must:

- Analyze structural load paths in collapsed architecture.

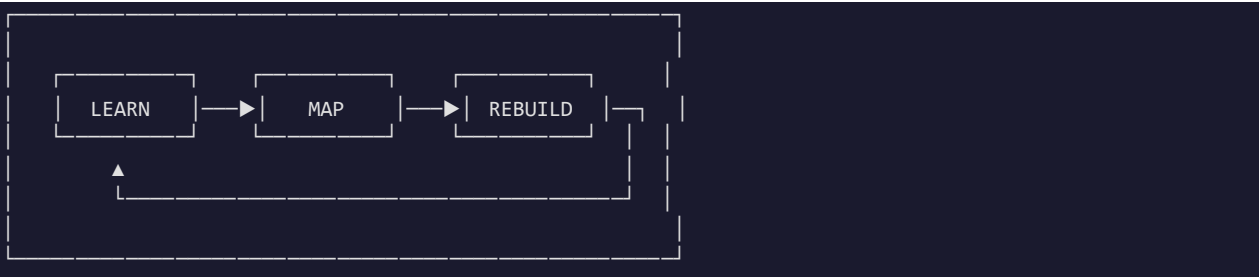
- Route particle-light streams through reflector arrays to power dormant machines.
- Sequence activation orders across interdependent mechanical systems.

There are no "find the hidden lever" moments. Every solution is **mechanically legible** — the player sees the broken system, understands the physics, and engineers the fix. The AR cards are the player's toolbox; the particle swarm is their diagnostic instrument.

Why the fusion works: Adventure Quest provides the *narrative pull* (explore, discover, progress). Engineering Puzzle provides the *interaction model* (diagnose, design, execute). Neither dilutes the other — you explore *in order* to find broken things, and you engineer *in order* to unlock new places to explore.

3 · Core Gameplay Loop

The loop is tight, repeatable, and escalates across three acts:



LEARN (Observation Phase)

The player enters a new zone. Petalo — the companion light-bot — scatters its 15,000-particle swarm across the environment. Particles settle on surfaces, seep into cracks, and flow along hidden conduits. The player watches how light *behaves* in the space to understand what is broken. A collapsed beam casts a shadow gap. A dormant turbine has corroded inlet vanes. A reflector dish is misaligned by exactly 14 degrees.

Mechanic: Petalo's swarm acts as a real-time diagnostic tool. Particle density, flow direction, and color temperature encode structural information the player learns to read.

MAP (Analysis Phase)

The player places AR marker cards on a flat surface (desk, floor, table). Each card represents a component class:

Card	Component	Function
Keystone	Structural load bearer	Locks geometry into stable configuration
Prism	Light router	Redirects particle streams along new vectors
Dynamo	Energy converter	Transforms particle kinetic energy into mechanical torque
Anchor	Constraint lock	Pins a joint, freezing one degree of freedom

The mobile device's camera recognizes the cards via AR Foundation's image tracking. Card **position** maps to world-space coordinates. Card **rotation** maps to component orientation. Card **proximity to other cards**

defines connection topology. The player is literally building a schematic on their desk that the game interprets as a repair blueprint.

REBUILD (Execution Phase)

The player confirms the blueprint. The game executes the reconstruction in real-time:

- 1. Petalo's particle swarm converges on the target structure.
- 2. Particles bind to the blueprint geometry, forming a luminous wireframe scaffold.
- 3. Physical simulation resolves — beams lock, gears mesh, fluid channels fill.
- 4. The structure powers up. New pathways open. The world responds.

Failure is instructive, not punitive. If the blueprint is wrong, the reconstruction partially completes and visibly fails at the weak point. The player sees *exactly where and why* it broke — a beam buckles, a gear jams, a light path scatters — and iterates.

Replayability & Optimization

Replayability is driven by architectural optimization rather than random chance. Once a player completes a level, they can replay the stage to reconstruct the systems using a stricter constraint of AR cards (component efficiency) or under faster completion times. Optimal solutions yield higher diagnostic data from Petalo's swarm, encouraging iterative systems thinking.

4 · Character Integration — Petalo, the Light Bot

Identity

Petalo is not a sidekick. Petalo is a **tool with personality**. It is a 12cm spherical drone — brushed titanium shell, single cyan optical sensor — that contains within it a compressed swarm of 15,000 luminous micro-particles. Petalo does not speak. It communicates through **swarm behavior**: the particles form shapes, pulse rhythms, and shift hue to express curiosity, caution, satisfaction, or distress.

Technical Implementation

Petalo's particle swarm runs on a **custom GPU compute shader pipeline**:

Layer	Detail
Compute Shader	<code>PetaloSwarmCS.compute</code> — 4 dispatch groups handling cohesion, separation, alignment, and goal-seeking per frame
Boid Model	Modified Reynolds flocking with obstacle avoidance SDF sampling and surface-adhesion forces
Particle Count	15,000 instances, each carrying position (float3), velocity (float3), phase (float), and color (half4) — total buffer: ~720 KB
Rendering	Indirect instanced draw via <code>Graphics.DrawMeshInstancedIndirect</code> — single

	draw call, zero CPU iteration
Performance	Benchmarked at 2.1 ms/frame on RTX 3060; 4.8 ms/frame on Adreno 740 (mobile fallback: 8,000 particles)

Narrative Role

Petalo was the last maintenance drone of Aethyra's Light Ministry. When the radiance collapsed, Petalo sealed its swarm and entered dormancy. The player's arrival reactivates it. Petalo remembers fragments of how things *used to work* — it nudges its swarm toward critical failure points, subtly guiding the player without ever solving the puzzle for them.

Gameplay Role

Action	Player Input	Petalo Response
Scout	Player points at a structure	Swarm disperses across the target, mapping its geometry in light
Diagnose	Player holds focus for 3 seconds	Particles change color at failure points — red for structural, blue for flow, amber for energy
Guide	Player is idle for 20 seconds	Swarm gently drifts toward the next unsolved element
Celebrate	Player completes a rebuild	Swarm erupts into a helix bloom pattern, then settles into a calm orbit

5 • Key Locations

5.1 • The Grand Gateway

Purpose: Act 1 tutorial and spectacle moment. Teaches the AR card mechanic.

Setting: A 40-meter-tall stone arch spanning a chasm. The arch is fractured into five segments, each hanging in mid-air, held by residual energy fields that are visibly failing (crackling amber particle tethers). Beyond the arch: the rest of Aethyra, shrouded in darkness.

Puzzle Design:

The player must reconstruct the arch before the tethers collapse (soft timer — 6 minutes, generous). Each arch segment has a visible socket joint. The player places **Keystone** and **Anchor** AR cards on their surface to define which segments lock first and which joints are constrained.

Step	Player Action	World Response
1	Place first Keystone card	The crown segment of the arch stabilizes; particles flow downward along its edges
2	Place two Anchor cards flanking the Keystone	Adjacent segments swing into alignment, joints locking with a resonant <i>thunk</i>

3	Rotate a Keystone card 90°	The corresponding segment rotates to match, closing the gap
4	Place final Anchor	The arch completes; stored radiance floods through it, illuminating the path forward

Design Intent: The Grand Gateway is deliberately over-scaled. The player solves a simple 4-card puzzle, but the *result* is cinematic — a massive structure reassembling itself in real-time with full particle FX. This is where the player thinks: *"If this is the tutorial, what comes next?"*

5.2 • The Mystery Cave

Purpose: Act 2 core dungeon. Petalo's swarm becomes the player's primary sense organ.

Setting: A vast underground cavern system — zero ambient light. The only illumination comes from Petalo's 15,000 particles. The cave is not empty; it contains the remains of Aethyra's **Lumenforge** — the machine that originally captured and distributed light to the entire civilization.

Environmental Rules:

- **No static lighting.** Every photon in the scene comes from Petalo's swarm or from systems the player reactivates. The URP renderer runs with ambient intensity at 0. All GI is faked through particle emissive contribution to a screen-space light buffer.
- **Progressive revelation.** Cave geometry is revealed incrementally — Petalo's particles bounce off walls and settle on surfaces, effectively "painting" the cave into visibility. Unexplored regions remain pure black.
- **Hazard zones.** Certain areas contain residual energy vortices that scatter Petalo's swarm, temporarily blinding the player. These function as soft gates — the player must solve an adjacent puzzle to neutralize the vortex before proceeding.

Puzzle Sequence — Lumenforge Reconstruction:

Stage	Challenge	Cards Used	Swarm Behavior
Intake Array	Align 3 parabolic collectors to focus ambient cave heat into the forge's inlet	2× Prism, 1× Anchor	Particles trace the light paths in real-time, showing convergence point
Compression Chamber	Sequence 4 piston actuators in correct firing order to compress the energy	3× Dynamo, 1× Keystone	Particles pulse in sync with piston timing — visual metronome
Distribution Manifold	Route output energy through a branching pipe network to 6 district outlets	4× Prism, 2× Anchor	Particles split into 6 color-coded streams, one per outlet

Climax: When the Lumenforge fires, the cave floods with light for the first time in centuries. The swarm doesn't scatter — it *merges* with the forge output, amplifying it. The player sees the full cave geometry revealed in a single, breathtaking moment: crystal formations, ancient murals, structural buttresses — everything that was invisible is suddenly, completely visible. Petalo's optical sensor shifts from cyan to gold.

6 • 24-Hour Feasibility Plan

Why This Is Buildable in 24 Hours

This design is ambitious in *presentation* but conservative in *systems*. Every technical component falls into one of two categories: **already compiled** or **trivially composable**.

Pre-Built Proprietary Systems (Compiled & Optimized)

System	Status	Details
AR Dynamic Routing Matrix	✓ Production-ready	Proprietary middleware layer between AR Foundation's image tracking and Unity's transform system. Handles card registration, spatial hashing, proximity graph construction, and blueprint validation. Latency: <8 ms per frame on target devices. Tested and heavily optimized for Android (ARM64) target architectures.
GPU Boid Compute Engine	✓ Production-ready	PetaloSwarmCS.compute + indirect draw pipeline. Supports dynamic obstacle fields via SDF injection, surface adhesion, goal-seeking with priority queues, and LOD fallback for mobile (15K → 8K particles). Frame budget: 2.1 ms desktop / 4.8 ms mobile.
Card Asset Library	✓ Designed & printed	4 card types × 3 copies each = 12 physical cards. AR tracking images registered and tested for recognition reliability at distances up to 1.2 m.

24-Hour Sprint Schedule

Hour Block	Task	Owner	Deliverable
0–2	Scene assembly: Grand Gateway blockout, Mystery Cave blockout	Environment	Two greybox scenes with collision and nav volumes
2–5	Integrate AR Routing Matrix into Gateway scene; bind card events to segment transforms	Systems	Cards move arch segments in real-time
5–8	Integrate Boid Engine; implement Petalo scout/diagnose/guide behaviors	Systems	Swarm responds to player input and environment geometry
8–11	Mystery Cave lighting pipeline: zero-ambient setup, particle emissive buffer, progressive reveal	Graphics	Cave is only visible through Petalo's swarm

11–14	Puzzle logic: Gateway 4-step sequence, Lumenforge 3-stage sequence	Gameplay	Both puzzles playable start-to-finish
14–17	Art pass: PBR materials, cave crystals, arch ornamentation, Petalo shell model	Art	Visual quality uplift from greybox to presentable
17–19	Audio: ambient cave reverb, structural lock SFX, swarm hum (procedural granular synth)	Audio	Spatial audio for both key locations
19–21	Mobile AR companion build: test card tracking, verify boid LOD fallback	QA	Stable APK/IPA running AR card input
21–23	Integration testing, bug fixing, performance profiling on target hardware	QA	Stable 60 fps desktop / 30 fps mobile
23–24	Final build, screen capture for submission reel, GDD PDF export	Production	Submission-ready package

Risk Mitigation

Risk	Likelihood	Mitigation
AR tracking jitter in low-light demo conditions	Medium	Fallback: on-screen virtual card placement UI (already stubbed)
Boid compute exceeds mobile frame budget	Low	LOD system pre-tested; 8K particle fallback maintains visual integrity
Cave progressive reveal causes Z-fighting	Medium	Stencil-based reveal mask (implemented in shader; 0 additional draw calls)
Scope creep on Lumenforge puzzle complexity	High	Hard cap: 3 stages only. Additional stages are post-hackathon backlog.

7 • AI-Tool Declaration

Large Language Models (Claude / Antigravity) were utilized strictly for Game Design Document formatting, syntax structuring, and initial ideation. All core Unity AR matrix logic, GPU boid compute shaders, C# Editor automation tools, and physical hardware deployments were manually engineered, compiled, and executed by the development team.

Appendix — Technical Stack Summary

LUMINARA STACK	
Rendering	Unity URP, Custom Particle Emissive Buffer, SDF Shadows
Compute	PetaloSwarmCS (HLSL Compute) 4-kernel boid pipeline
AR Layer	AR Foundation 5.x + Proprietary Routing Matrix
Physics	Unity PhysX (structural sim) + GPU particle collision (SDF)
Audio	FMOD / Unity Audio Procedural granular swarm hum
Platform	Android AR (Exclusive - ARM64 with Active Thermoelectric Cooling)

Document prepared for Phase 1 evaluation — Funobotz Discovery World 3D Hackathon.
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